BUJI — Personal AI Trainer

Project Proposal & Development Blueprint

SmartUp Learning Ventures Pvt. Ltd.

Prepared by: Pydart Intelli Corp | Date: April 2026 | Version 1.0

Executive Summary

Buji is a first-of-its-kind Personal AI Trainer purpose-built for SmartUp Learning's online tutoring ecosystem. Unlike generic chatbots, Buji is a deeply integrated, continuously learning AI companion that knows every student personally — their academic history, classroom behavior, exam performance, attention patterns, strengths, weaknesses, and learning style.

Buji transforms SmartUp from a tutoring platform into an **intelligent learning partner** — available 24/7, infinitely patient, and increasingly personalized with every interaction.

Timeline: 2 months (4 phases) **Current Status**: Working prototype deployed (basic chat, exam data, memory persistence)

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1. The Vision

> *"Every student at SmartUp will have their own AI trainer who knows them better than any tutor ever could — one that never sleeps, never forgets, and gets smarter every day."*

What Makes Buji Different

Generic AI Chatbot	Buji — Personal AI Trainer
Generic answers from public data	Answers grounded in the student's **actual** data
No memory between sessions	**Persistent memory** — remembers goals, struggles, preferences

Generic AI Chatbot	Buji — Personal AI Trainer
Cannot teach or explain	**Step-by-step academic coaching** with curriculum alignment
Text-only interaction	**Voice, image, document** — multimodal interaction
Same experience for everyone	**Hyper-personalized** — adapts to each student's learning style
No integration with platform	**Deep integration** — classroom, exams, attendance, fees, everything
Static capability	**Continuously learning** — improves with every conversation and data point

Core Philosophy

1. **Data-Driven Personalization** — Every response is informed by real academic data
2. **Curriculum-First Intelligence** — Aligned to CBSE, ICSE, ISC, and State Boards
3. **Proactive, Not Reactive** — Buji reaches out, not just responds
4. **Multi-Stakeholder Value** — Serves students, parents, teachers, and administrators
5. **Privacy by Design** — Each user sees only their own data, fully compliant with data protection norms

2. Current State (Prototype)

The working prototype validates Buji's core architecture. Currently deployed at `smartuplearning.online`:

What's Built

Feature	Status
Chat interface (text + file upload)	Live
Server-side comprehensive data fetching (16 data sections)	Live
Persistent memory across sessions	Live
Full exam question-level data (Q&A, options, correct answers)	Live
Per-session AI monitoring breakdown	Live
Attendance, fees, credits data awareness	Live
Image analysis (vision model)	Live
PDF document analysis	Live
Parent dashboard integration	Live
Fault-tolerant data pipeline (section isolation)	Live
Auto memory extraction (goals, weak subjects, study habits)	Live

What's Planned (This Proposal)

The complete Buji system expands from this foundation into a full-spectrum AI trainer with 40+ features across 8 capability domains.

3. Complete Feature Specification

3.1 Academic Coaching Engine

The heart of Buji — a teaching AI that explains, not just answers.

#	Feature	Description
1	**Step-by-Step Problem Solving**	When a student gets a question wrong, Buji explains *why* the correct answer is right and *why* their answer was wrong — with mathematical/scientific reasoning
2	**Doubt Resolution**	Students can type or photograph any question from textbook/homework. Buji solves it with full working, aligned to their board & curriculum
3	**Concept Explanation**	Ask "What is an equivalence relation?" and Buji explains at the student's level, referencing topics from their actual classes
4	**Multi-Step Solution Builder**	For complex problems (integration, chemical equations, derivations), Buji builds solutions step-by-step with option to expand each step
5	**Practice Question Generator**	"Give me 5 more questions like Q2 from my last exam" — Buji generates similar-difficulty questions on weak topics
6	**Topic Deep Dive**	Student selects any topic from their curriculum → Buji generates a structured mini-lesson with examples, practice problems, and a quick quiz
7	**Formula & Theorem Reference**	Instant recall of relevant formulas, theorems, and derivations — curriculum-aligned (CBSE/ICSE/State)
8	**Exam Preparation Coach**	Before exams: Buji creates a personalized study plan based on weak areas, time available, and exam syllabus
9	**Handwriting & Image Analysis**	Photograph handwritten work → Buji reads it, checks for errors, suggests corrections
10	**Multi-Subject Expertise**	Mathematics, Physics, Chemistry, Biology, English, Hindi, Social Science — board-specific content

3.2 Performance Analytics & Insights

Buji as the student's personal data analyst.

#	Feature	Description

#	Feature	Description
11	**Exam Score Analysis**	Detailed breakdown of every exam — per-question analysis, topic-wise accuracy, time management insights
12	**Strength & Weakness Mapping**	Continuously updated profile: "You're strong in Relations & Functions but weak in Equivalence Classes"
13	**Trend Detection**	"Your Math scores improved 15% over the last 3 exams" or "Your attention scores are declining in Chemistry"
14	**Comparative Insight**	Performance relative to batch averages (anonymized) — "You scored above 80% of your class in this exam"
15	**Attention & Behavior Reports**	Natural language summary of AI monitoring data — "You were distracted 4 times in Tuesday's Math class, mainly in the last 20 minutes"
16	**Weekly Performance Digest**	Auto-generated weekly summary: attendance, exams, attention, homework status — delivered proactively
17	**Parent-Friendly Reports**	Simplified insights for parents: "Tishnu attended 4/5 classes this week. Math exam score improved."

3.3 Study Planning & Scheduling

Buji as the student's personal academic planner.

#	Feature	Description
18	**Smart Study Planner**	Input: available hours, exam dates, weak subjects → Output: day-by-day personalized study schedule
19	**Revision Reminders**	Spaced repetition engine — "It's been 5 days since you studied Organic Chemistry. Time for revision!"
20	**Homework Tracker**	"You have 2 pending homework assignments. Chemistry is due tomorrow."
21	**Session Preparation Tips**	Before each class: "Today's Math class covers Integration by Parts. Here's a quick refresher on substitution..."
22	**Goal Setting & Tracking**	"I want to score 90% in Physics" → Buji creates milestones, tracks progress, adjusts plan

3.4 Multimodal Interaction

Beyond text — natural, accessible communication.

#	Feature	Description

#	Feature	Description
23	**Voice Input (Speech-to-Text)**	Students can speak their questions — especially useful for younger students or during study sessions
24	**Voice Response (Text-to-Speech)**	Buji reads answers aloud — helpful for revision while commuting or before sleep
25	**Image-Based Q&A**	Photograph textbook pages, whiteboard notes, or handwritten problems → Buji analyzes and responds
26	**PDF Analysis**	Upload previous year papers, study material, or notes → Buji summarizes, creates flashcards, or generates practice tests
27	**Multi-Language Support**	Primary: English & Hindi. Buji detects language automatically and responds in the same language
28	**LaTeX Math Rendering**	Mathematical expressions rendered beautifully — fractions, integrals, matrices displayed correctly

3.5 Proactive Intelligence

Buji doesn't wait to be asked — it reaches out.

#	Feature	Description
29	**Class Reminders**	"Your Chemistry class starts in 30 minutes. Don't forget to join!" (push notification + chat message)
30	**Low Attendance Alerts**	"You've missed 3 Chemistry classes this month. Your attendance is at 60%. Let's fix this!"
31	**Fee Reminders**	"You have 2 overdue invoices. Would you like me to guide you to the payment page?"
32	**Credit Warnings**	"You have only 6 Math credits remaining out of 50. Ask your parents to renew."
33	**Post-Exam Follow-up**	After every exam: "You scored 40% on your Math exam. Want me to go through the questions you got wrong?"
34	**Motivational Nudges**	"You've improved 12% in Chemistry this month! Keep going, Tishnu!"
35	**Inactivity Check-In**	If a student hasn't chatted in 5+ days: "Hey Tishnu, haven't heard from you! How are your studies going?"

3.6 Classroom Integration

Buji connected to the live classroom experience.

#	Feature	Description

#	Feature	Description
36	**Post-Class Summary**	After each live class: Buji generates a summary of topics covered, key formulas, and homework assigned
37	**In-Class Quick Reference**	During live sessions, students can ask Buji quick questions without leaving the classroom
38	**Recording Notes**	When class recordings are available: Buji generates topic-wise timestamps and key takeaways
39	**Teacher Feedback Relay**	If teachers leave comments on homework or exams, Buji notifies and explains the feedback

3.7 Parent Experience

Different Buji for parents — focused on monitoring and communication.

#	Feature	Description
40	**Child Progress Dashboard**	"How is my child doing?" → Comprehensive updates across all subjects, attendance, behavior
41	**Attendance Alerts**	Instant notification when child misses a class or joins late
42	**Exam Results Notification**	"Your child scored 85% in today's Mathematics exam" — with detailed breakdown
43	**Fee Management**	"What fees are pending?" → Exact amounts, due dates, payment links
44	**Teacher Communication**	"Can you tell the Math teacher my child will miss class tomorrow?" → Routes to appropriate channel
45	**Weekly Parent Report**	Auto-generated digestible report: attendance, exam scores, behavior, teacher comments

3.8 Learning & Memory System

Buji's persistent intelligence layer.

#	Feature	Description
46	**Conversation Memory**	Remembers what was discussed — "Last time you asked about Integration. Want to continue?"
47	**Student Profile Learning**	Learns study habits, preferred explanation styles, weak topics, goals — builds over time
48	**Adaptive Difficulty**	Adjusts explanation complexity based on the student's demonstrated understanding level
49	**Knowledge Graph**	Internal map of what each student knows and doesn't know — used to personalize every response

#	Feature	Description
50	**Feedback Loop**	"Was this explanation helpful?" → Student feedback fine-tunes Buji's approach for that student

4. Technical Architecture

4.1 Technology Stack

Component	Technology	Purpose
LLM Core	Groq Cloud API (Llama 3.3 70B)	Text reasoning, academic coaching, analysis
Vision AI	Groq Cloud API (Llama 3.2 11B Vision)	Image analysis, handwriting recognition
Speech-to-Text	Web Speech API / Whisper	Voice input processing
Text-to-Speech	Web Speech API / ElevenLabs	Voice response generation
Database	PostgreSQL 15	Student data, exam records, memory storage
Cache	Redis	Response caching, rate limiting, hot memory
Framework	Next.js 16 (App Router)	API routes, SSR, real-time UI
Frontend	React + Tailwind + shadcn/ui	Chat interface, quick actions
Notifications	BullMQ + SMTP + WhatsApp API	Proactive alerts and digests
Math Rendering	KaTeX	LaTeX equation display in chat
Curriculum Data	Custom DB (CBSE/ICSE/State)	Board-aligned content generation

5. AI Training & Data Pipeline

5.1 How Buji Learns

Buji uses a **three-layer learning system**:

```
Layer 1: REAL-TIME CONTEXT (per-request)
├── 16 data sections fetched live from PostgreSQL
├── Full exam Q&A, attendance, monitoring, fees
├── Always current — no stale data
└── ~8,000-20,000 chars of structured context

Layer 2: PERSISTENT MEMORY (per-student)
├── Extracted insights from every conversation
├── Goals, weak subjects, study habits, preferences
```

```
  ├── Learning style, emotional state, family context
  ├── Grows over time — Buji gets smarter per student
  └── Stored in chatbot_memory table

  Layer 3: CURRICULUM KNOWLEDGE (global)
  ├── CBSE, ICSE, ISC, State Board syllabi
  ├── Subject-wise topic trees with prerequisites
  ├── Formula banks, theorem databases
  ├── Previous year question patterns
  └── Updated annually for syllabus changes
```

5.2 Continuous Training Cycle

```
┌──────────────────────────────────────────────────┐
│              CONTINUOUS LEARNING LOOP            │
│                                                  │
│  Student Interacts ──► Data Captured             │
│         │                    │                   │
│         ▼                    ▼                   │
│  Memory Extracted     Feedback Logged            │
│         │                    │                   │
│         ▼                    ▼                   │
│  Profile Updated      Response Quality Tracked   │
│         │                    │                   │
│         ▼                    ▼                   │
│  Next Response ◄── Personalization Applied       │
│                                                  │
└──────────────────────────────────────────────────┘
```

5.3 Data Sources for Buji Intelligence

Data Source	Volume	Updates
Student profiles	1 per student	On profile change
Exam results + full Q&A	20+ per student	After each exam
AI monitoring events	50+ events per session	Real-time during class
Attendance records	1 per session per student	After each session
Homework submissions	Varies	On submission
Fee & payment records	Monthly	On payment
Session credits	Per enrollment	On usage
Conversation history	10 recent turns	Per chat
Persistent memory	30+ entries per student	Per conversation
Curriculum data	CBSE/ICSE/State	Annual update

6. Role-Wise Buji Capabilities

For Students

Capability	Example Interaction
Doubt solving	*"Explain the difference between permutation and combination with examples"*
Exam review	*"Go through my wrong answers from today's exam and teach me"*
Study planning	*"Create a study plan for my Physics exam next Monday"*
Performance check	*"How am I doing in Chemistry compared to last month?"*
Homework help	*"Help me solve this integration problem" (uploads photo)*
Quick reference	*"What's the formula for centripetal acceleration?"*
Practice questions	*"Give me 5 tough questions on electromagnetic induction"*
Motivation	*"I'm feeling stressed about exams"*
Platform queries	*"When is my next Math class?"*
Fee queries	*"How many credits do I have left?"*

For Parents

Capability	Example Interaction
Daily check-in	*"How was my child's day at SmartUp?"*
Attendance query	*"Has Tishnu been attending all classes?"*
Exam results	*"How did my child do in today's exam?"*
Behavior report	*"Is my child paying attention in class?"*
Fee check	*"What fees are pending?"*
Weekly summary	*"Give me this week's report"*
Teacher interaction	*"Who is teaching Mathematics to my child?"*
Credits	*"How many session credits are remaining?"*

For Teachers (Future Phase)

Capability	Example Interaction
Class insights	*"How is my Batch A performing overall?"*
Student flags	*"Which students need attention in Math?"*
Exam analytics	*"What topics were students weakest in?"*
Attendance trends	*"Show me attendance trends for this month"*

For Batch Coordinators (Future Phase)

Capability	Example Interaction
Batch overview	*"Give me today's session status"*
Student alerts	*"Which students have declining performance?"*
Session summary	*"Summarize today's monitoring alerts"*

7. Development Phases & Timeline

Phase 1: Foundation (Month 1, Week 1–2) — *Elevate the Prototype*

Task	Details
Academic Coaching Engine	Step-by-step problem solving, doubt resolution with curriculum alignment
Enhanced Memory System	Knowledge graph per student, adaptive difficulty, feedback loop
Improved Chat UI	Quick action buttons, message reactions, copy/share, LaTeX rendering
Intent Classification	Route queries to specialized handlers (academic, admin, analytics)
Error Recovery	Graceful handling of all edge cases, fallback responses
Response Quality	Improve prompts, test across subjects, validate accuracy

Deliverable: Production-ready academic coaching for students

Phase 2: Intelligence (Month 1, Week 3–4) — *Make Buji Smart*

Task	Details
Strength/Weakness Engine	Topic-level mastery tracking from exam + homework data
Trend Analysis	Week-over-week, subject-wise performance trends
Practice Question Generator	Generate curriculum-aligned questions targeting weak areas
Study Planner	Smart scheduling based on exam dates, weak subjects, available time
Revision System	Spaced repetition reminders based on forgetting curve
Comparative Analytics	Anonymized batch-level comparisons

Deliverable: Intelligent analytics and personalized study planning

Phase 3: Multimodal & Proactive (Month 2, Week 1–2) — *Beyond Text*

Task	Details
Voice Input	Speech-to-text integration for hands-free interaction
Voice Response	Text-to-speech for audio explanations

Task	Details
Advanced Image Analysis	Handwriting recognition, diagram analysis, whiteboard capture
Proactive Notifications	Class reminders, fee alerts, exam follow-ups, inactivity nudges
Post-Class Summaries	Auto-generated class summaries with topic timestamps
WhatsApp Integration	Buji responses mirrored to WhatsApp for parents

Deliverable: Multimodal interaction + proactive engagement system

Phase 4: Scale & Polish (Month 2, Week 3–4) — *Production Excellence*

Task	Details
Parent Buji	Full parent experience — progress reports, alerts, fee management
Response Caching	Redis cache for frequent queries, reduced API costs
Rate Limiting	Per-user rate limits to prevent abuse
Analytics Dashboard	Usage metrics, satisfaction tracking, popular queries
Load Testing	Performance optimization for 500+ concurrent users
Documentation	Admin guide, teacher guide, student onboarding materials
UAT & Bug Fixes	User acceptance testing across all roles and devices

Deliverable: Production-ready, fully-featured Buji AI Trainer

8. Competitive Advantage

8.1 Market Comparison

Feature	Byju's AI	Toppr AI	Vedantu AI	SmartUp Buji
Real student data awareness	No		No	Yes
Live classroom integration	No		Partial	Yes
AI monitoring behavior data	No		No	Yes
Exam Q&A-level coaching	No	Basic	No	Yes
Persistent memory	No		No	Yes
Parent AI assistant	No		No	Yes
Proactive outreach	No		No	Yes
Voice interaction	No		No	Yes
Fee & admin queries	No		No	Yes
Curriculum-aligned practice	Yes		Partial	Yes

8.2 Cost Breakdown

> **Scope & Pricing Note:** This estimate covers development and integration with the SmartUp online portal only. Integration with any offline ERP system and custom AI model training are **not included** in this scope. As features scale and training data volume grows, costs may increase. All revisions will be agreed upon in advance.

Category	Amount (₹)	%	Details
AI Development & Engineering	2,40,000	55.8%	Core AI logic, context pipeline, memory system, intent routing, academic coaching engine, practice generation, analytics
Frontend & UI/UX	60,000	14.0%	Chat redesign, voice UI, quick actions, LaTeX rendering, mobile optimization, parent interface
Multimodal Integration	40,000	9.3%	Voice I/O, advanced image processing, handwriting recognition, PDF intelligence
Proactive System	30,000	7.0%	Notification engine, class reminders, auto-reports, WhatsApp integration, scheduling
Testing & QA	25,000	5.8%	Academic accuracy validation, load testing, cross-device testing, edge case handling
AI API Costs (6 months)	20,000	4.7%	Groq Cloud API usage for development and initial production period
Documentation & Training	15,000	3.5%	Admin docs, user guides, onboarding materials, video walkthroughs
TOTAL	**4,30,000**	**100%**	

> **Optional Add-on — 3D Character Development**: A custom animated 3D avatar for Buji (idle, talking, thinking, celebrating states) is available as a separate engagement. Cost quoted on request based on character design complexity and animation requirements. Not included in the ₹4,30,000 estimate above.

Recurring Costs (Post-Launch)

Item	Estimated Monthly Cost
Groq API (at scale)	₹3,000–8,000/month (usage-based)
Server resources (incremental)	₹0 (uses existing SmartUp servers)
Maintenance & updates	Included in platform SLA

8.3 Unique Selling Points

1. **Zero-Setup Personalization**: Buji knows the student from day one — no quizzes or onboarding required

2. **Classroom-Connected**: Only AI that has live classroom behavior data (attention, engagement, tab switches)
3. **Full Academic Lifecycle**: Covers pre-class prep → live class → post-class review → exam prep → results analysis
4. **Parent Partner**: Keeps parents informed without manual teacher effort
5. **Continuous Learning**: Gets smarter with every interaction — not a static chatbot

8.4 Investment & Payment

> **Note on Pricing:** The cost of ₹4,30,000 is an **estimated figure** for development and integration with the SmartUp online portal only. Integration with any offline ERP system and custom AI model training are **not included** in this estimate. As the number of features, students, and training data grows, the actual cost may increase. Any scope changes will be communicated and agreed upon before execution.

Item	Details
Estimated Total Cost	₹4,30,000 (Four Lakhs Thirty Thousand Only)
Timeline	2 months from project kickoff
Payment — Advance (50%)	₹2,15,000 — on project kickoff
Payment — Mid (30%)	₹1,29,000 — on Phase 2 completion
Payment — Final (20%)	₹86,000 — on final delivery & UAT sign-off
Warranty	30 days post-launch bug fixes included

9. Success Metrics & KPIs

Engagement Metrics

Metric	Target (3 months)	Target (6 months)
Daily active users (students)	40% of enrolled	70% of enrolled
Avg. messages per student/week	10+	20+
Returning users (weekly)	60%	80%
Parent adoption rate	30%	50%

Academic Impact Metrics

Metric	Target
Exam score improvement (Buji users vs non-users)	+15% average
Homework completion rate	+25% improvement
Attendance improvement for flagged students	+20%
Student-reported confidence increase	70% positive

10. Risk Mitigation

Risk	Probability	Impact	Mitigation
LLM gives incorrect academic answers	Medium	High	Multi-layer validation: curriculum DB cross-check, confidence scoring, "I'm not sure" fallback, teacher review flagging
API cost escalation at scale	Medium	Medium	Response caching, rate limiting, smaller model for simple queries, batch processing
Student over-dependence on AI	Low	Medium	Encourage self-solving first, Socratic method (ask questions back), teacher notification for repeated help requests
Data privacy concerns	Low	High	Strict role isolation, no cross-student data leakage, DPDPA compliance, audit logging
LLM provider downtime	Low	Medium	Fallback to cached responses, queue mechanism, multi-provider support
Low adoption by students	Medium	Medium	Gamification (streaks, badges), proactive engagement, teacher endorsement, onboarding nudges

11. Terms & Acceptance

Delivery Terms

- **Estimated Total Cost**: ₹4,30,000 (Rupees Four Lakhs Thirty Thousand Only)
- **Timeline**: 2 months from project kickoff
- **Payment Schedule**:
  - 50% advance on project kickoff — ₹2,15,000
  - 30% on Phase 2 completion — ₹1,29,000
  - 20% on final delivery & UAT sign-off — ₹86,000
- **Warranty**: 30 days post-launch bug fixes included

> The above cost is an estimate for SmartUp online portal integration only. Due to the scale of features and the volume of data required, labour and infrastructure costs may increase. Any revisions will be discussed and approved in advance.

What's Included

- Complete Buji AI system (50 features as specified)
- Student, Parent, and Admin interfaces
- Voice input/output integration
- Proactive notification system
- WhatsApp integration for parents
- Deployment to production servers
- User documentation and training materials

What's Not Included

- Integration with any offline ERP or third-party school management system
- Custom AI model training or fine-tuning on proprietary datasets
- Hardware or infrastructure procurement
- Features outside the scope defined in Section 3

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